

Meru मेरु

A private AI workspace on machines your company manages.

For a CTO or CIO weighing a pilot, and for a platform team weighing a managed offering. Facts, measurements and gaps, as of this release.

v0.4.19 · October 2026 · aarora79.github.io/meru

github.com/aarora79/meru
Scan for the code

THE QUESTION

The data AI helps with most is the data a company least wants to send out

- **A hosted assistant reads each question on its provider's servers**, with the files, mail and notes attached to it.
- **Each one is a data processor** to assess, contract with and monitor, and its terms can change.
- **Meru takes the other path.** Open-weight models run on the employee's own machine, beside the data, and no model provider is in the path.

WHAT MERU IS

An assistant that runs as a program on each person's machine

One daemon, three clients

`merud` keeps the models warm and owns the data. Meru.app, `meru chat` in a terminal, and `meru run --json` for scripts talk to it over a local socket.

Local models

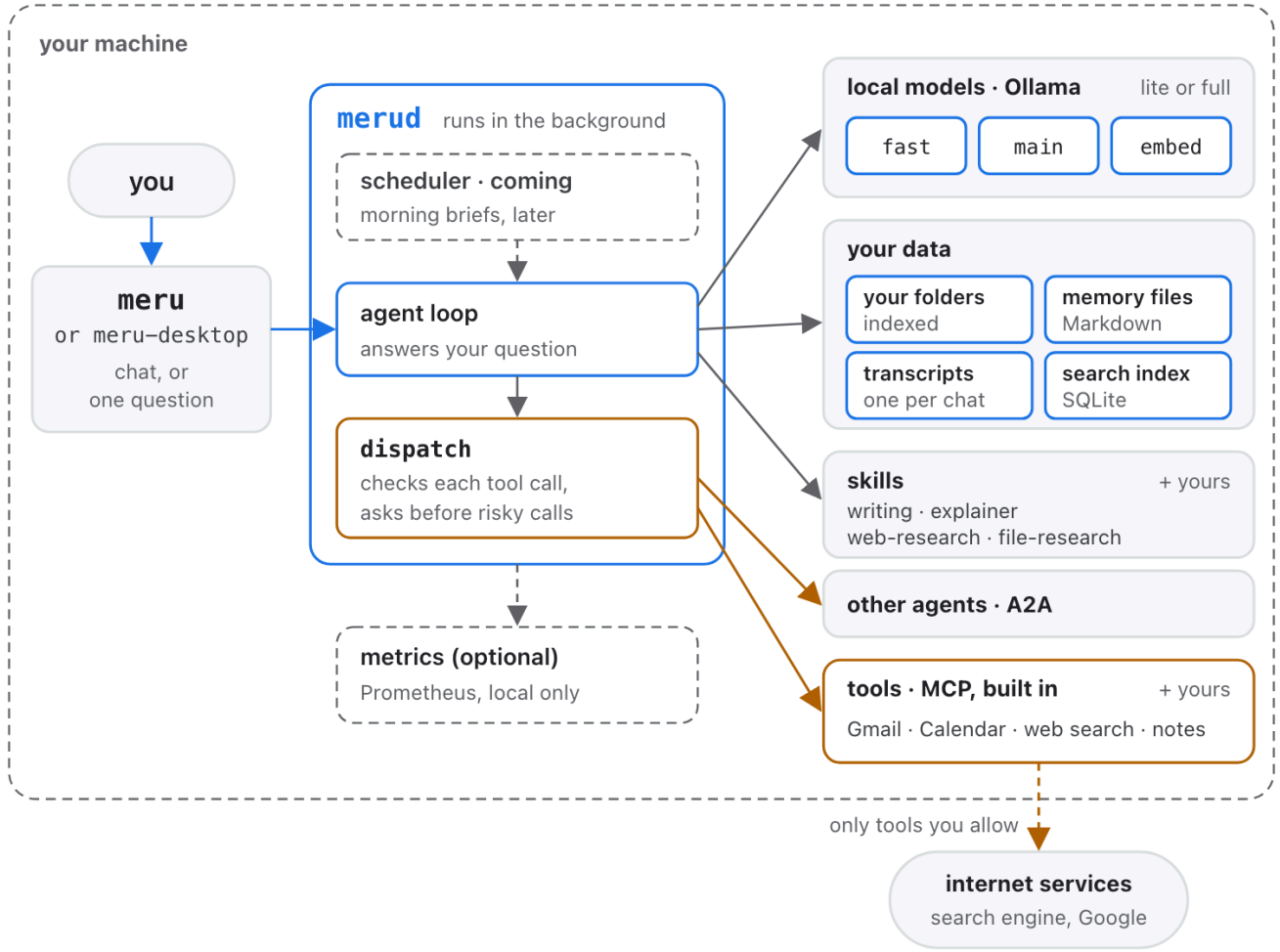
Open-weight models through Ollama on 127.0.0.1. The code has no path to a cloud model, and a test in CI fails the build if one appears.

The person's own data

Their folders, mail and calendar, notes in Obsidian or Apple Notes, the web through a search engine they run, and their voice through Whisper.

Apache License 2.0. Written in Go. 19 releases since the first commit on 23 September 2026. macOS on Apple silicon first; Linux supported; Windows untested.

THE BOUNDARY



Everything inside the dashed line runs on the employee's machine

The models, the search index, the chat history and the memories stay there. Only tools the configuration allows reach outside.

DATA FLOW

What leaves the machine, and what doesn't

Data	Where it goes
Questions and answers	Nowhere. The models run in Ollama on the same machine.
Files and notes	Indexed into a SQLite file on the machine; read live, never copied out.
Telemetry, crash reports, update checks	None. Metrics go to a Prometheus the company runs on the same machine.
Web searches	The search words go to the engines the company's own SearXNG asks.
Web pages	Public addresses only. Meru asks first for an address no search or question gave, unless the user trusted that website.
Mail and calendar	Read through the user's own Google account, when they connect it.

CONTROLS

Controls a security team can check

Control	How Meru does it
Tools	Deny by default. A new tool server contributes nothing until config names its tools. Each tool is Off, Ask or Allow; sending, deleting or changing asks every time.
Network	Connects only to its own machine, except to servers marked remote and to public pages it reads; refuses private addresses at connect time.
Secrets	API keys in one file only the user can read; kept out of the model, the logs and the history.
Supply chain	Every program Meru installs is pinned by version and SHA-256 or image digest; CI signs a record of how its speech server was built.
Enforcement	Policy tests in CI fail the build on a cloud model, a second path to a tool, or a forbidden import.

AUDIT

Every tool call leaves a record on the machine

- **One path for every tool call.** MCP servers, other agents, local commands and built-in tools all go through `dispatch`, which checks the allowlist, asks for approval, and logs the call.
- **Two records.** A `tool_calls` row with the tool, time, duration, outcome and approval, and the session's transcript, a plain JSON Lines file.
- **Incognito keeps the audit row.** An incognito chat keeps no transcript, and its rows keep no arguments or results, but each call is still counted.
- **Metrics and traces** in OpenTelemetry, to a collector on the same machine only. Prompt and answer text stay out of traces unless config turns capture on.

MEASURED

Answer quality and speed by Mac memory

Mac memory	Answer model	Passed	First word, median	Whole answer, median
under 32 GB	`MiniCPM5-2B`	68%	11.6 s	13.1 s
32 to 47 GB	`gemma4:26b-a4b-it-qat`	80%	17.6 s	18.3 s
48 to 63 GB	`qwen3.6:35b`	89%	5.6 s	10.4 s
64 GB or more	`qwen3.6:35b-a3b-mxfp8`	89%	4.1 s	6.5 s

150 results per model set, 3 runs of 50 questions about the author's own files, notes, mail and calendar, which aren't published. Times include routing, search and tool rounds. Your numbers on your data will differ. Source: docs/benchmarks/results.md, 28 September 2026.

SIZING

What the hardware has to hold

Memory, as Ollama loaded it

- `qwen3.6:35b` with a 262,144-token window: 28.8 GB
- The same model at 32,768 tokens: 23.4 GB
- `MiniCPM5-2B` , the fast model: 2.8 GB
- The search model: 0.7 GB

What follows

- A 64 GB Apple silicon Mac runs the full set with room to spare.
- Under 32 GB, Meru runs the 2B model and passes fewer questions.
- Meru sizes each answer model's window to fit an eighth of memory.
- The answer model wrote 84 to 93 tokens a second on an M4 Max.

ROLLOUT TODAY

What it takes to put Meru on a machine

- **A Mac installer** on a disk image sets up Ollama, web search, folders and `merud` in eleven steps in a window. A one-line installer does the same in a terminal.
- **One config file per person**, `~/.meru/config.toml`: the models, the folders, the tools and what each may do.
- **Updates are manual.** Meru never checks for them, so the company decides when a new version reaches each machine.
- **Files are the source of truth.** The database can be deleted and rebuilt from the transcripts, memories and config.

GAPS

What Meru does not do today

For a fleet

- No central console to push config or policy
- No single sign-on or directory link
- No central log collection: metrics and traces stay on the machine by design
- One person per machine; no shared server

For the project

- Pre-alpha, v0.4.19
- One maintainer; the first outside contribution merged in October 2026
- Scheduled jobs come in v0.5
- Windows builds, untested

FOR A PLATFORM PROVIDER

What a managed offering around Meru would have to add

Fleet setup and config

Install, update and push `config.toml` with each team's allowlists, as the company's device management already does for other programs.

A model catalog

Open-weight models chosen per hardware tier, pinned by digest and served from a mirror the company controls.

Hardware

Apple silicon Macs or GPU machines with enough memory per person, on desks or in the company's own account; the boundary holds wherever the machine is.

Connectors and audit

MCP servers for the company's own systems, and a collector the company runs on each machine to gather the audit rows Meru keeps there.

The Apache License 2.0 allows this. Meru's core stays local by design: no cloud model, no telemetry, deny by default.

FOR A CTO OR CIO

A four-week pilot with ten people

Week	What happens	What to look at
1	Pick ten people with Apple silicon Macs of 32 GB or more. Install, add their work folders, connect one system.	Install time; the controls table checked against the config
2 and 3	They use it for real work: questions on their files, mail, notes and the web.	The Usage screen: questions, routes, answer times, tool calls by outcome
4	Security reviews the `tool_calls` rows and the network rules; the ten say what worked.	Pass or fail against the company's own criteria



Built in the open

The code, the design documents and every decision are in the repository.

We're happy to take contributions: see the **issue list**. Questions and pilots start there too.

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